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Electrochemical acid–base generators for decoupled carbon management†

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Carbon dioxide capture and management are critical technologies for achieving carbon neutrality and mitigating the impacts of global warming. One promising approach for decarbonization involves electrochemical generation of concentrated acid and base. This effectively decouples the carbon capture-release process from the electrochemical cell, avoiding the kinetic limitations associated with reactions involving CO_2. Designing an electrochemical acid–base generator with high current efficiency and low energy cost is challenging. Following investigations of the crossover rates of protons and hydroxide ions through ion-exchange membranes, we designed a multichambered electrochemical cell for generating weak acid and strong base, which significantly suppressed acid–base crossover. By equipping the center chamber with a serpentine flow field, we achieved acid–base production at high concentrations (>1 M) and high coulombic efficiency (>95%) while maintaining relatively low energy costs. With this device, we demonstrated carbon management examples of simulated flue gas capture, direct air capture, and green production of slaked lime, as one step toward green cement production. The key components of the prototype can be adapted for use in other electrochemical cell designs, ensuring high efficiency in concentrated acid–base generation in other application scenarios.

Broader context

In the effort to combat climate change, technologies that capture and manage carbon dioxide (CO_2) are essential. Among these, a promising method uses electricity to generate acid and base, which helps capture and release CO_2 without the usual energy-intensive steps required for these reactions. This electrochemical approach can be fine-tuned to capture CO_2 from various sources, including industrial flue gas and even the air. Our research explores how to improve this technique by designing a specialized electrochemical cell that produces both a strong base and a weak acid while minimizing energy waste. This cell design uses a multi-chamber structure and specific flow patterns to create high-concentration acid and base efficiently. By reducing the "crossover" or mixing between acid and base, our system maintains high efficiency with lower energy requirements, offering a solution for efficiently generating concentrated acid and base for usage outside of the electrochemical cell. Beyond capturing CO_2, this system can aid in producing materials like "slaked lime," a key ingredient for making greener cement. The modular nature of the components means they can be tailored for various applications, making this an energy-efficient approach to carbon management for a more sustainable future.

The escalating threat of global warming has spurred concerted efforts to mitigate the concentration of greenhouse gases, particularly carbon dioxide, in the earth's atmosphere.[1] Specifically, techniques that can collect pure CO_2 from dilute sources (flue gas, air, and ocean) are the first steps before any downstream carbon utilization[2] or storage.[3] Various methodologies have been explored to address this pressing environmental challenge.[4] CO_2 can be captured from flue gas using liquid amine absorbents by applying a thermal swing, but this approach is infeasible for direct air capture (DAC) due to ventilation of amines.[5] For the more challenging DAC application, thermal swing[6] or moisture swing[7] on porous solid adsorbents has been used. Metal–organic frameworks,[8,9] amine-decorated silica,[10] and alkaline polymers[11] have all been developed for DAC, though these are still hampered by limited thermal and chemical stability. Meanwhile, cycling methods involving the thermal decomposition of calcium carbonate ($CaCO_3$) into calcium oxide (CaO) present a scalable approach for DAC,[12] but they suffer from high energy consumption.[13]

Electrochemical methods, without requiring the activation of the whole substrate as in thermal swing can leverage

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renewable energy sources and facilitate decentralized operations with potentially greater energy efficiency.[14] Electrochemical amine regeneration for the carbon capture cycle has been realized through mediated regeneration[15,16] and direct carbon dioxide reduction of the post-capture solvent.[17] One notable technique involves redox-active materials that have reversible binding with CO_2.[18,19] Taking quinones as an instance, CO_2 can chemically bind with reduced quinones and be released upon the oxidation of reduced quinones.[20,21] Most of the reduced forms of these molecules, however, are sensitive to oxygen, which is prevalent in carbon capture environments. Moreover, the inherent correlation between the reduced molecules' binding affinity with CO_2 and reactivity with O_2 makes the development of such molecules for DAC challenging.[22] Aqueous redox-active molecules undergoing proton-coupled electron transfer (PCET) enable effective pH-swing cycles during charge and discharge, facilitating CO_2 absorption by reaction with hydroxide and emission; they can operate at high current densities.[23,24] Most electrochemically activated molecules for CO_2 capture are susceptible to reversible chemical oxidation by atmospheric oxygen, requiring further development of rebalancing methods.[25]

Electrochemical acid–base generators, by separating generated acid and base from redox mediators, have the potential to overcome the challenge of oxygen sensitivity. The chlor-alkali process and the water splitting driven salt splitting process are examples of electrochemical acid–base generators, but they have low energy efficiency.[26–31] Similar generators using the iodine-alkali process lower the working voltage.[32] Other acid–base generators have been developed including non-PCET redox mediators with bipolar membranes (BPMs)[33] and PCET mediators[34] such as oxygen[35] or hydrogen[36–40] with multichambered cells for acid–base generation. Water dissociation driven salt splitting using BPMs, though suffering from a very high ohmic resistance due to multiple center chambers and membranes, has been used historically and already is at industrial scale for acid–base generation.[33,41] However, there are still several desirable properties for carbon capture applications that most electrochemical acid–base generators have not yet fully met. First of all, for any electrochemically-driven carbon capture process, a lower ohmic resistance is crucial for lowering the capital cost and operational cost (Fig. S1, ESI†). Ohmic resistance is the major voltage consumer under high current density operation.[39] Furthermore, for DAC, decoupling CO_2 capture and release from the electrochemical cell can avoid the drawbacks of slow kinetics related to CO_2 reactions[35] and CO_2 bubble formation in the center-chambers in the cell,[38] by creating design flexibility for engineering CO_2 mass transport and kinetics (*e.g.*, utilization of carbonic anhydrase[42]). The strategy of decoupling CO_2 reactions from the electrochemical cell can further open up other carbon management related applications requiring acid and base.[43] Finally, a remaining goal of the generator based on a decoupled process is achieving high current efficiency across a wide range of current densities. Important parameters like current density, energy cost, current efficiency of CO_2 capture, CO_2 source concentration, CO_2 outgas concentration and dead volume are correlated; hence achieving only a few good numbers at one specific operation condition is not very helpful. Only when all of the parameters are co-optimized during the cycled or steady-state operation, can an electrochemical cell be potentially practical. A comparison between reported electrochemical acid–base generators is shown in Table S1 (ESI†).

In this work, we present a three-chamber electrochemical cell that used a center chamber with a serpentine flow field, as well as an ion buffer system to manipulate the charge carrier type, in order to achieve a higher current efficiency (Fig. 1a). The cell configuration comprised an anode chamber, a central chamber, and a cathode chamber, each separated by cation exchange membranes (CEMs). We utilized cycled hydrogen gas as an example of a low overpotential PCET redox mediator for acid–base generation. The hydrogen oxidation reaction (HOR: $H_2 - 2e^- \rightarrow 2H^+$) and the hydrogen evolution reaction (HER: $2H_2O + 2e^- \rightarrow H_2 + 2OH^-$) occurred at the anode and cathode, respectively. Only a small initial quantity of H_2 was necessary to initiate operation, without further consumption. Specifically, hydrogen was introduced into the anode chamber where it undergoes oxidation on a gas diffusion electrode (GDE) to yield protons. These protons traversed the anode CEM, binding with weak acid anions (A^-), accumulating weak acid (HA) that can be collected. Concurrently, maintaining charge neutrality, sodium ions migrated through the cathode CEM into the cathode chamber, where the hydrogen evolution reaction occurred, leading to the production of metal hydroxide (MOH) and enabling the collection of base (Fig. 1b). Consequently, generated acid and base could facilitate carbon capture ($2MOH + CO_2 \rightarrow M_2CO_3 + H_2O$; $M_2CO_3 + CO_2 + H_2O \rightarrow 2MHCO_3$) and release upon subsequent acid–base recombination ($HA + M_2CO_3 \rightarrow MA + MHCO_3$, $HA + MHCO_3 \rightarrow MA + H_2O + CO_2$), respectively. Using weak acid buffering couples, the charge carrier exchange occurring in the center chamber was the key process for maintaining high current efficiency under high current density. With improved cell structure and operation conditions, we were able to achieve high current density acid–base generation for decoupled carbon capture with a relatively low energy cost. The generated high-concentration acid and base can be used outside of the electrochemical cell, avoiding solid clogging, turbulent resistance or slow reaction kinetics during utilization inside the cell.

Acid–base crossover in the generator

We systematically studied the crossover of acid and base in the electrochemical cell to gain insights into high-efficiency designs. To assess the extent of current efficiency loss attributable to the crossover of hydroxide and protons (Fig. 1b), it is imperative to accurately quantify the crossover flux through each membrane.

We first studied hydroxide crossover in the acid–base generator independently. We implemented a two-chamber, one-membrane cell configuration, featuring a hydrogen evolution reaction (HER) cathode alongside a non-PCET neutral anode, wherein the ferrocyanide oxidation reaction (FOR: $Fe(CN)_6^{4-} - e^- \rightarrow Fe(CN)_6^{3-}$) occurs. FumaSep® E610-K was used as the

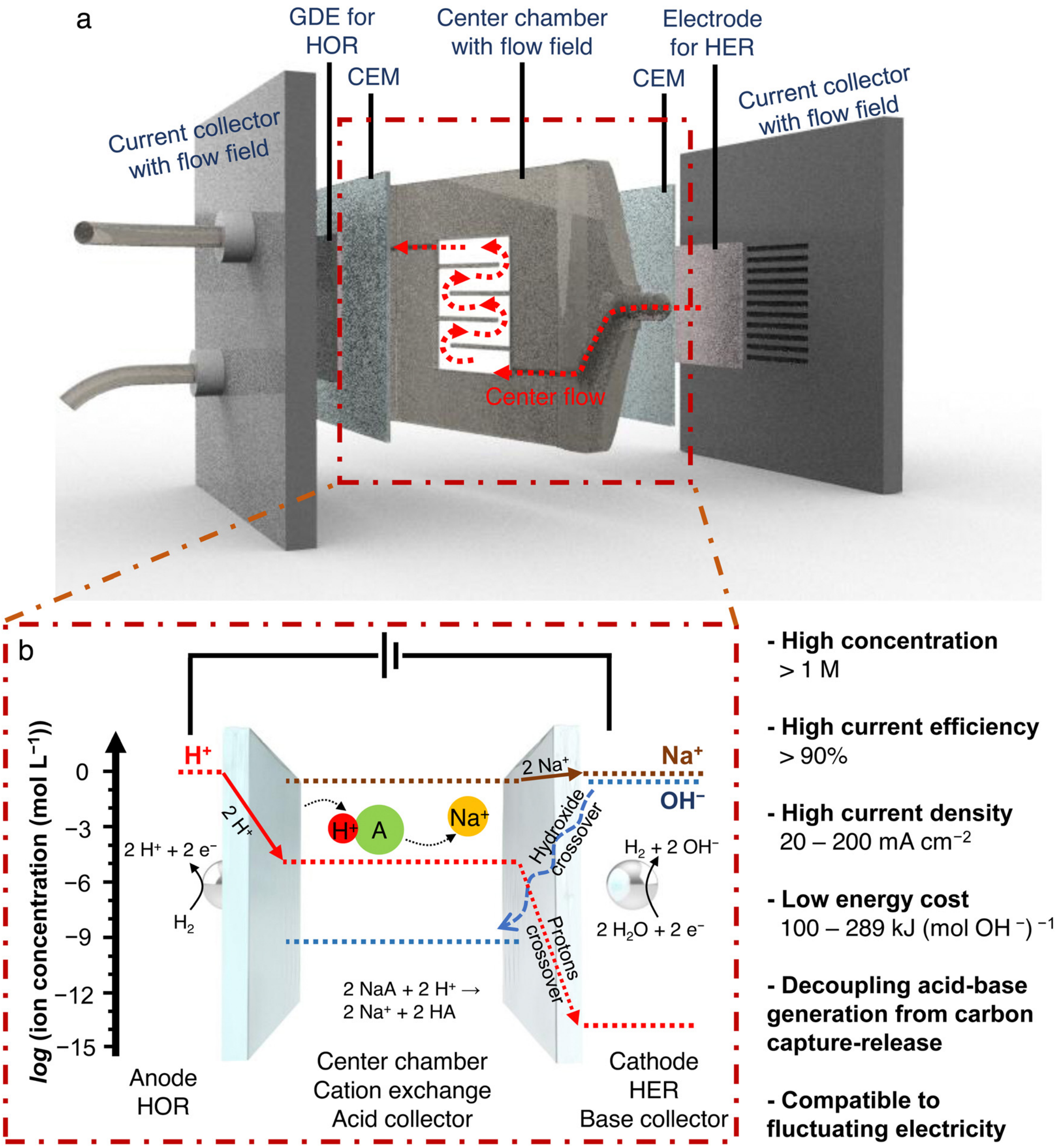

Fig. 1 Cell structure with technique comparison. (a) Structure of the three-chamber two-membrane acid–base generator with components and flow labeled. (b) Schematic of the ion transport and concentration in the acid–base generator. Flux of protons (left solid arrow) and sodium ions (right solid arrow) are major charge carriers. Crossover of protons and hydroxide ions (dashed arrows) are only minor charge carriers penalizing the current efficiency.

CEM in the cell, mimicking the hydroxide crossover from the HER side in the acid–base generator (Fig. 2a). Employing an *in situ* pH sensor within the anode chamber facilitated the measurement of pH variations resulting from hydroxide crossover from the cathode chamber. To maintain stability in the concentration of NaOH throughout the testing period, the volume of the solution in the cathode chamber exceeded that in the anode chamber by a factor of 4. Different concentrations of NaOH cycled in the catholyte and current densities were introduced to evaluate their impact on hydroxide crossover (Fig. S2 and Table S2, ESI†). By correlating pH values with concentration and factoring in the volume of the cathode chamber, we derived the time-dependent molarity of hydroxide ions (Fig. S3 and Table S3, ESI†). Notably, the hydroxide concentration exhibited a linear increase over time, enabling the determination of the average crossover flux *via* linear fitting of the curve. As shown in Fig. 2b, a higher concentration of hydroxide ions in the source side and a higher current density exacerbates the crossover flux. The results were described and fitted by the Nernst–Plank equation using a two-factor regression model with terms for diffusion and electromigration (Fig. S4, ESI†).

Subsequently, to assess proton crossover, we employed a three-chamber, two-membrane cell similar to the structure of the acid–base generator. In this cell, iodine reduction ($I_3^- + 2e^- \rightarrow 3I^-$) was utilized as a non-PCET neutral cathode reaction.

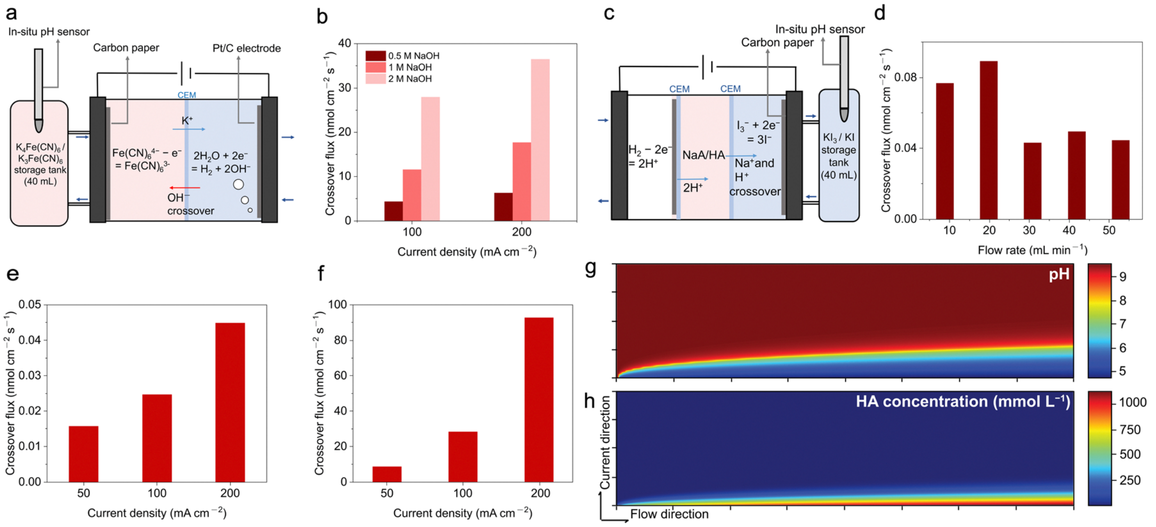

Fig. 2 Hydroxide and proton crossover test. (a) Schematic of a two-chamber one-membrane cell used for hydroxide crossover test. (b) Crossover flux of hydroxide under different current densities and concentrations of sodium hydroxide used in the cathode side. (c) Schematic of a three-chamber two-membrane cell used for proton crossover test. (d) Proton crossover flux with various flow rates of 2 M NaOAc in the center chamber. The current density is 200 mA cm^{-2}. (e) Proton crossover flux with various current density applied, cycling 2 M NaOAc in the center chamber with a flow rate of 53 mL min^{-1}. (f) Proton crossover flux with various current density applied, cycling 1 M NaOAc + 1 M HOAc in the center chamber with a flow rate of 53 mL min^{-1}. Simulation of the distribution of (g), pH and (h), HA in the center chamber (horizontal length: 11.18 cm, vertical length: 1.0 mm), with a pK_a of HA equaling 4.75. The figures are ideal top views from the orientation of the center chamber in Fig. 1. The direction of the flow is from left to right. Bottom boundaries indicate the anode membrane where protons enter. The top boundaries indicate the cathode membrane where sodium ions exit, with minor proton crossover.

HOR was used as the anode reaction, donating protons into the center chamber, as shown in Fig. 2c. The volume of the solution in the center chamber was augmented to maintain solution composition stability, with an *in situ* pH sensor deployed in the cathode chamber to monitor pH fluctuations resulting from proton crossover.

We ran the cell with a solution of 2 M sodium acetate (NaOAc) in the center chamber, corresponding to the initial condition during operation. By analyzing the accumulated molarities of protons in the catholyte under different flow rates in the center chamber (Fig. S5 and Table S4, ESI†), we quantified the proton crossover flux and observed a slight increase under lower flow rate (<30 mL min^{-1}) of the center chamber electrolyte (Fig. 2d) at a current density of 200 mA cm^{-2}. We attribute the increase of crossover to weaker washout of accumulated HOAc and protons, and decreased NaOAc supply. Additionally, for a fixed flow rate of 53 mL min^{-1} in the center chamber, proton crossover increased with higher current density (Fig. 2e, Fig. S6a and Table S5, ESI†). A significant increase in proton crossover was observed when the solution composition in the center chamber was changed from 2 M NaOAc to a mixture of 1 M NaOAc and 1 M HOAc (Fig. 2f, Fig. S6b and Table S6, ESI†). The results indicate that the buffer strength of 1 M NaOAc and 1 M HOAc is diminished compared to 2 M NaOAc, causing more protons to directly cross two membranes. Crossover of protons may also be due to HOAc ionization and HOAc crossover as a whole molecule (Fig. S7, S8 and Table S7, ESI†). Potential crossover of acetic acid can be avoided by using polymerized weak acids (*e.g.*, polyacrylic acid), at the expense of increasing the ionic resistance (Fig. S9, ESI†).

We modeled acid crossover under advection, assuming a straight uniform center chamber (Fig. 2g and h, Note S1 and Fig. S10, ESI†), to capture the crossover behavior observed experimentally. Upon entry of protons into the central chamber, the weak acid (HA) is generated. Given the pK_a of acetic acid (4.75), the local proton concentration remains low, with most protons being sequestered by acetate ions within the chamber. Moreover, the concentration of sodium ions in the chamber is approximately five orders of magnitude higher than that of protons, enabling sodium ions to predominantly serve as charge carriers, thereby minimizing proton crossover through the cathode CEM. These observations offer insights for improving the performance of multi-chamber electrochemical acid–base generators systems.

Characterization of the acid–base generator

To lower the influence of crossover and increase the concentration of sodium ions as charge carriers in the center chamber, we chose 20 mL of 2 M NaOAc as an initial input solution, pumped through the center chamber of a 5 cm^2 cell. We cycled the electrolyte for the center chamber and cathode to accumulate weak acid and base, expecting to generate weak acid and strong base with a concentration exceeding 1 M. During cell operation, we titrated the concentration of generated base periodically to evaluate the differential current efficiency corresponding to each short time interval by dividing the increased molarity of acid and base with the number of electrons passing through the circuit. This differential current

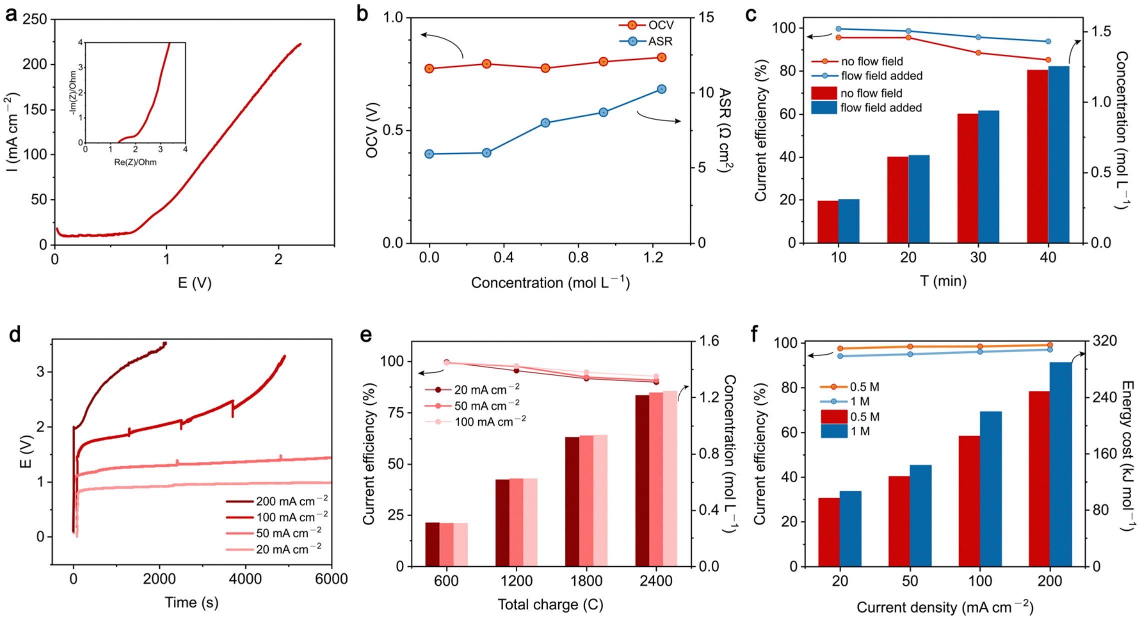

Fig. 3 Performance of weak acid, strong base generator. (a) LSV curve and EIS of a 5 cm^2 acid–base generator at the initial operation condition. (b) OCV and ASR of the generator corresponding to the concentration of HOAc in the center chamber during operation. (c) Differential current efficiency and concentration of accumulated acid and base during operation under a current density of 200 mA cm^{-2}. Differential current efficiency is calculated over a short interval as explained further in the main text. (d) E–t curve during operation under different current densities. (e) Comparison of differential current efficiency and concentration of generated acid and base under different current densities. Horizontal axis is total charge passing through the circuit, meaning the amount of generated acid and base should be the same at each data point under different current densities, thus the current efficiency is comparable. (f) Round trip current efficiency and energy cost of obtaining 0.5 M and 1 M of acid and base using the generator.

efficiency reflects the relationship between acid–base crossover and the temporal electrolyte composition. The overall integrated current efficiency was calculated by dividing molarity of the total acid and base with the total moles of passed electrons, after reaching the specified concentration of accumulated acid and base. The polarization of the acid–base generator was analyzed using linear sweep voltammetry (LSV) with 2 M NaOAc cycled in the center chamber, as shown in Fig. 3a. Electrochemical impedance spectroscopy (EIS) was also performed to evaluate the resistance of the cell. The cell demonstrated a current density of about 200 mA cm^{-2} under 2.0 V with a high-frequency area-specific resistance (ASR) of 5.95 Ω cm^2. During the operation of the generator, the replacement of sodium ions with protons in the center chamber alters the ratio between NaOAc and HOAc, thereby increasing the resistance. Fig. 3b illustrates the change of high-frequency ASR with an increasing concentration of HOAc during operation. When HOAc reached 1.2 M (NaOAc = 0.8 M), high-frequency ASR increased from 5.95 Ω cm^2 to 10.25 Ω cm^2 because of the higher ohmic resistance of the center chamber solution (Fig. S11, ESI†), emphasizing the critical role of electrolyte composition in improving the energy efficiency of the generator. Although the pH in the center chamber and catholyte changed during operation, the open circuit voltage (OCV) of the cell remained stable. An OCV of about 0.8 V corresponds to the thermodynamic voltage of generating pH 0 acid and pH 14 base, which are the local pH values of the HOR GDE anode and the HER alkaline cathode, respectively.

To maximize current efficiency, a more uniform advection in the center chamber is desirable. We designed a serpentine flow field within the center chamber to facilitate the flow in the center. Fig. S8 (ESI†) shows the comparison of flow through a center chamber with and without a flow field. The flow field enables the solution to distribute evenly in the window, thereby increasing ion exchange efficiency and avoiding stagnant areas where sodium is depleted and replaced by HOAc, increasing the local ohmic resistance and crossover of acids. Consequently, with the addition of a flow field, the acid–base generator showed improved differential current efficiency (Fig. 3c), due to a more even flow distribution (Fig. S12, ESI†).

We further investigated the performance of the 5 cm^2 cell under different current densities. The cell voltage against operation time is plotted in Fig. 3d. A 20 mL solution of 2 M NaOAc was applied and cycled in the center chamber at t = 0. The increase in cell voltage over time is due to the increasing ohmic resistance as charge-carrying sodium ions are depleted and replaced by HOAc. Thus, the voltage increase over time is more pronounced under higher current densities. To make the current efficiencies and generated acid–base under different current densities comparable, we collected and titrated the acid–base when a certain molarity of charge passed through the circuit. Under all current densities, the cell promises a differential current efficiency above 95% at the beginning and remains above 90% after generating acid–base with a concentration higher than 1 M (Fig. 3e). The cell with higher current density demonstrated a slightly higher current efficiency, though crossover flux of acid and base was previously found to be larger. This is because the time the cell spends in order to achieve the same molarity of accumulated acid and base is shorter. Integrated current efficiency and energy cost of the acid–base generator are related to the target concentration of accumulated acid and base. Upon reaching 1 M acid and base (Fig. 3f), the integrated current efficiency of the acid–base generator under all current densities exceeds 94%. Specifically, the cell demonstrates operational viability below 1.0 V when subjected to a current density of 20 mA cm^{-2}, boasting an integrated current efficiency of 94.2% and demonstrating an integrated energy cost of 106 kJ mol^{-1}. The integrated energy cost increased to 220 kJ mol^{-1} at 100 mA cm^{-2}, which is still lower than most of the acid–base methods under the same current density.[32] Better alkaline HER and acidic HOR catalysts should lower the energy cost. For example, utilizing Pt–Ru alloy rather than Pt for alkaline HER lowered the cell voltage by 15.2% under 200 mA cm^{-2} (Fig. S13, ESI†). Ohmic resistance is still the major voltage consumer under high current densities, calling for the development of better membranes with lower resistance and higher selectivity, as well as thinner center chambers (Fig. S14 and S15, ESI†). With a better air contactor, a lower concentration of generated base can be used for kinetically acceptable DAC. If only accumulating 0.5 M acid and base, the integrated current efficiency and energy cost can be further reduced (98% current efficiency, 180 kJ mol^{-1} energy cost at 100 mA cm^{-2}).

Applications of the acid–base generator

In order to validate the functionality of the acid–base generator in practical scenarios, we employed alternative power sources in lieu of the conventional electrochemical workstation. Specifically, we utilized a home-made solar cell station (150 cm^2) to power the 5 cm^2 acid–base generator with a hydrogen gas loop (Fig. 4a and Fig. S16, ESI†). Although the solar cell voltage fluctuated over time, it exhibited an average output voltage of approximately 1.7 V and a current of approximately 0.35 A during a cold and overcast December day in Boston (Fig. 4b). To assess the performance of the acid–base generator under the alternative power source, we conducted analyses of integrated current efficiency. Remarkably, despite operating under fluctuating voltage conditions, the acid–base generator exhibited a current efficiency exceeding 90%. These results underscore the robustness and versatility of the acid–base generator, confirming its efficiency across diverse power generation platforms. Additionally, we used a flow battery to energize the acid–base generator, thereby simulating operating conditions without directly connecting to the grid (Fig. S17, ESI†). Similarly, the flow battery, utilizing Br_3^-/Br^- in the posolyte and H_2DPPEAQ/DPPEAQ (((9,10-dioxo-9,10-dihydroanthracene-2,6-diyl)bis(oxy))bis(propane-3,1-diyl))bis(phosphonic acid) in the negolyte,[44,45] yielded a stable voltage of around 1.3 V with an average current of 0.23 A, exhibiting a current efficiency of acid–base generation exceeding 92% due to a relatively more stable voltage output (Fig. 4c). These findings validate the potential applicability of the acid–base generator in practical settings, offering a promising pathway toward sustainable and decentralized energy generation technologies without grid electricity supply.

The utilization of the generated acid and base offers promising avenues for carbon management processes, encompassing carbon

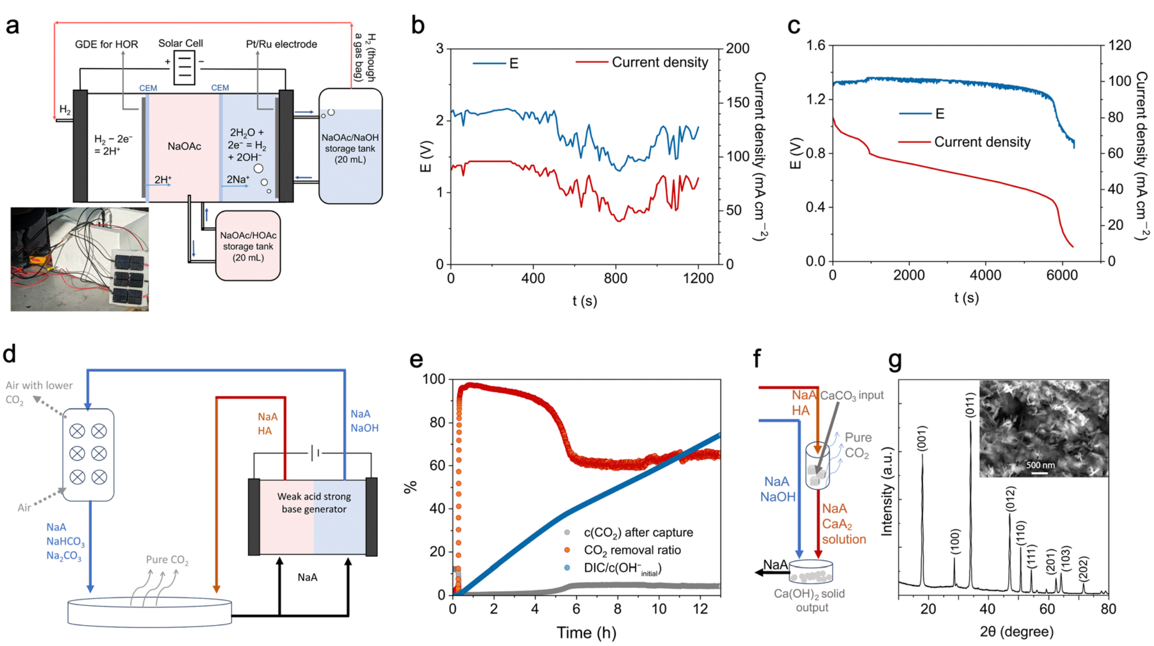

Fig. 4 Applications of the acid–base generator. (a) Digital pictures and schematic of the acid–base generator powered by a home-made solar panel. Components are labeled accordingly (b), *E–t* and *I–t* curve of the cell during the solar cell powered generator operation. (c) *E–t* and *I–t* curve of the cell during the flow battery powered generator operation. (d) Schematic of the carbon dioxide capture-release cycle. (e) CO_2 concentration in the exhausted gas after capture, CO_2 removal ratio and DIC/c($OH^-_{initial}$) during the carbon capture for simulated flue gas. (f) Schematic of the $Ca(OH)_2$ production process. (g) XRD of the $Ca(OH)_2$ prepared using generated acid and base; inset: SEM image.

capture from simulated flue gas and air (Fig. 4d). Following 40 minutes of operation under a current density of 200 mA cm^{-2}, we acquired a NaOH solution and a weak acid (HOAc) solution with a concentration of 1.2 M. This solution was subsequently employed for carbon management applications.

The base solution we obtained was subsequently employed for capturing CO_2 from simulated flue gas containing 10% CO_2, 3% O_2, and 87% N_2. The simulated flue gas was bubbled through the base solution, subsequently entering the CO_2 sensor for post-capturing gas characterization. Fig. 4e illustrates the temporal evolution of post-capture CO_2 concentration (in %, at 1 atm), along with the corresponding CO_2 removal efficiency. The removal efficiency (CO_2 concentration in gas before capture divided by CO_2 concentration in exhausted gas after capture) consistently exceeds 90% throughout the initial 3-hour period. We use the increased concentration of dissolved inorganic carbon (DIC) after capture divided by the initial concentration of hydroxide ions ($c(OH^-{}_{\text{initial}})$) to quantify the utilization of hydroxide ions. The DIC/$c(OH^-{}_{\text{initial}})$ reached 70% after 12 h. The decreasing removal efficiency over time is caused by the depletion of hydroxides, because lower pH reduces the capture reaction rate. The generation of NaOH, as a strong base presenting a strong driving force for the absorption of CO_2, is more favored for carbon capture from a more diluted source such as air than for capture from flue gas. For direct air capture (DAC), after 2 days of exposing 20 mL of the obtained base in a 100 cm^2 container to air with natural flow, the DIC/$c(OH^-{}_{\text{initial}})$ reached about 48% (corresponding to reaction $CO_2 + 2OH^- = CO_3{}^{2-} + H_2O$ going nearly to completion), with notable evaporation of water. With a better air contactor, the absorbing rate, removal ratio and utilization of hydroxide may be further improved.[12] Contaminations such as SO_x and NO_x will not poison the catalysts, but their accumulation requires further treatment (Fig. S18, ESI†).

During cyclic operation, a certain amount of the dissolved inorganic carbon (DIC) remains unreleased during acidification. This 'baseline DIC' acts as a buffer by consuming the generated hydroxide ions during acid–base generation which, in turn, decreases system efficiency. Once the system reaches steady cyclic operation, however, 100% of the CO_2 captured in each cycle is released in the same cycle. To quantify this behavior, we measured the initial amounts of generated acid and base to determine the extent of CO_2 capture and release, thereby accounting for this baseline DIC. After mixing with generated acid, about 95% of the captured CO_2 was released (Note S2, ESI†). The amount outgassed was confirmed by adding the generated acid into a sealed container with the after-capture solution under vigorous stirring. The volume of the outgassed CO_2 was measured. It is also worth noting that if CO_2 release is directly coupled within the cell (using after-capture solution in the center chamber), the resistance and acid–base crossover rates increase and become unstable (Fig. S19, ESI†).

As another example of carbon management using acid and base, we explore the application of the generated products in green slaked lime production. The first step of traditional cement manufacturing involves the calcination of calcium carbonate to obtain calcium oxide, followed by its reaction with water to yield calcium hydroxide. This existing process is energy-intensive and produces substantial amounts of carbon dioxide. In response, we propose a more sustainable method[43] using the generator, where the acid reacts with calcium carbonate to yield a calcium salt solution. This reaction produces an outgas composed solely of CO_2, which is suitable for direct utilization or storage. Following this, the calcium salt solution reacts with a base to precipitate high-purity calcium hydroxide (Fig. 4f). The dissolving and precipitation process are high in efficiency (Note S3, ESI†). Fig. 4g shows the X-ray diffraction (XRD) pattern and scanning electron microscopy (SEM) of the synthesized $Ca(OH)_2$ nanosheets, highlighting the purity. These results demonstrate the potential of the generator to directly transform green electricity into low carbon industry, providing sustainable and affordable pathways for CO_2 capture and slaked lime production.

The long-term stability of electrochemical CO_2 capture is a key factor in determining its feasibility for industrial-scale deployment. Our system benefits from the decoupling of electrochemical and chemical processes, which mitigates many stability concerns associated with conventional capture methods. The absence of direct exposure of catalyst to air, CO_2, NO_x, or SO_x extends the lifetime of the electrochemical components, reducing maintenance requirements. However, several challenges must be addressed for large-scale implementation.

From a deployment perspective, integrating this approach with existing industrial CO_2 sources requires careful consideration of pretreatment, electrolyte handling and regeneration logistics. The presence of contaminations such as NO_x and SO_x, while not directly impacting the electrochemical cell, still influences the downstream chemistry and may require additional base supplementation to maintain system performance. Addressing these challenges through process optimization could enable widespread adoption of electrochemical carbon management.

Designing high efficiency acid–base generators

The concept of utilizing weak acid to increase the current efficiency of the acid–base generator for decoupled carbon management can improve the performance of other electrochemical acid–base generators. A few benefits come along with the utilization of weak acid buffer layer. The first outcome is a simpler cell structure, resulting in a lower resistance. Compared to acid–base generators achieving strong acid and base, one less center chamber and one less anion exchange membrane (AEM) are used in the acid–base generators (Fig. S20 and S21, ESI†). Furthermore, the utilization of a weak acid buffer solution allows the generation of concentrated acid ($>$1 M) without dramatically increasing acid–base crossover, which is challenging for strong acids due to fast crossover of protons through AEMs.[46] The utilization of a weak acid to decouple CO_2 releasing reactions from the electrochemical cell also avoids the precipitation of solid in the electrochemical cell or bubbles in the center chamber, which is generally problematic.

For future engineering toward such applications, key considerations of capture-release integrated energy efficiency and long-term system practicability should drive a convincing techno-economic analysis. For the three-chamber hydrogen-driven weak acid–base generator, a lower electricity cost and a current density of 50–200 mA cm^{-2} can possibly lead to a commercially sound carbon capture levelized cost (Fig. S22, ESI†). This positions electrochemical CO_2 capture as a potentially competitive alternative to conventional methods, particularly in scalable scenarios where flexible and modular deployment is desired.

To demonstrate the broad applicability of our strategy, we tested the efficiency of various acid–base generators utilizing weak acid buffers. A BPM based water dissociation acid–base generator driven by cycled ferro/ferricyanide oxidation/reduction is presented in the schematic (Fig. 5a). At varying current densities, the *E*–*t* curves were plotted in Fig. 5b, dominated by the ohmic resistance of the cell (Fig. S23a, ESI†). The overall current efficiency of the acid–base generation under 200 mA cm^{-2} is 94% (Fig. 5c). Note, however, that the efficiency diminishes under low current density conditions due to the low water dissociation efficiency in the BPM, caused by ionic crossover.[47] This problem could be ameliorated by the development of higher efficiency BPMs with a lower resistance and crossover.

Using oxygen molecules from the air as redox mediators avoids gas looping engineering.[35] In the oxygen-driven cell (Fig. 5d and Fig. S23b, ESI†), the anodic and cathodic reactions are the oxygen evolution reaction (OER) and oxygen reduction reaction (ORR) respectively. The cell maintained a commendable current efficiency of approximately 92% even at low current densities, though it can hardly achieve 200 mA cm^{-2} under a reasonable voltage due to the high overpotential of OER and the mass transport limitation of atmospheric oxygen for the ORR (Fig. 5e and f). This can possibly be overcome with better catalysts and GDE structures.[35] The water-splitting cell (Fig. 5g and Fig. S23c, ESI†) exhibits a simpler design, comprising a single membrane and two chambers. It yields a decent cell voltage, generating green hydrogen along with acid and base (Fig. 5h). Current efficiencies of the cell exceeded 91% under various current densities (Fig. 5i), supporting the robustness of our design principle in achieving simultaneous, high-efficiency acid–base generation.

Conclusion

With a low-overpotential, hydrogen looping, two-CEM electrochemical cell, we realized highly efficient generation of weak

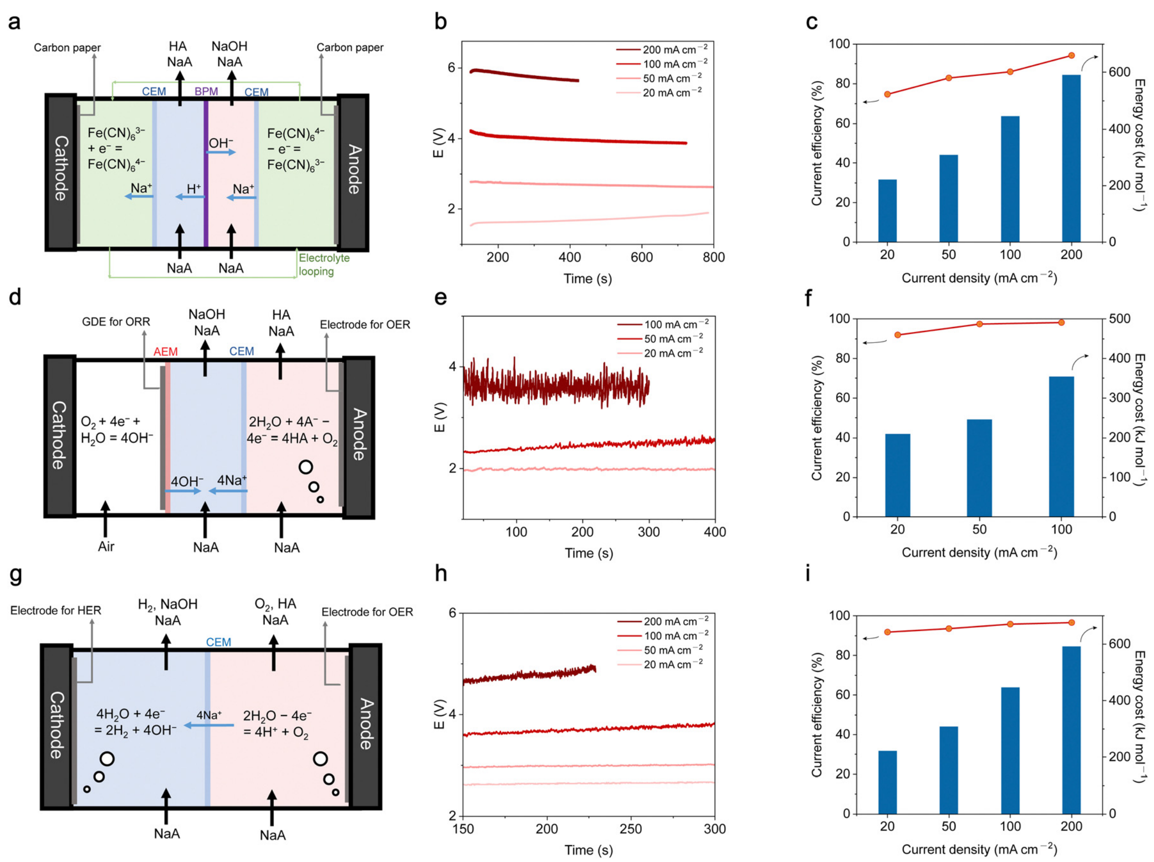

Fig. 5 Other acid–base generators utilizing a weak acid buffer chamber. (a) Schematic of a BPM based four-chamber three-membrane acid–base generator driven by non-PCET reversible redox reactions. (b) *E*–*t* curve of cell (a) under different current densities. (c) Integrated current efficiency and energy cost to obtain 1 M acid–base by using cell (a). (d) Schematic of three-chamber two-membrane acid–base generator driven by OER and ORR. (e) *E*–*t* curve of cell (d) under different current densities. (f) Integrated current efficiency and energy cost to obtain 1 M acid–base by using cell (e). (g) Schematic of acid–base generator driven by water splitting. (h) *E*–*t* curve of cell (g) under different current densities. (i) Integrated current efficiency and energy cost to obtain 1 M acid–base by using cell (g).

acid and strong base. The design reduced the ohmic resistance as the main energy consumer under high current density operation and suppressed acid–base crossover as the major contributor to current efficiency loss. The introduction of a serpentine flow field for the center chamber further increased the current efficiency by improving the uniformity of advection in the cell. The flow field also reduced flow resistance and thickness of the center chamber, leading to lower ohmic resistance and lower operational cost of multi-chamber cells. With various designs of acid–base generators, we showed that the utilization of a weak acid buffer system is a robust strategy with broad applicability to achieve current efficiencies exceeding 90% and to simplify the cell structure, thus bringing convenience to cell designs that feature simpler structures and lower resistance. The utilization of polymerized soluble weak acids can further lower the crossover and increase the system robustness though showing a higher resistance. Adding extra supporting salts (*e.g.*, NaCl) can potentially further lower the cell resistance and proton crossover, leading to a lower energy cost.

The generated weak acid and strong base can be of high concentration and enable decoupled carbon management applications. The utilization of weak acid and strong base in simulated flue gas carbon capture, DAC and green slaked lime production is thermodynamically and kinetically practical, without involving a problematic gas phase in the center chamber or solid precipitation in the electrochemical cell. Decoupling the chemical reactions outside of the electrochemical cell allows better reactor designs to accelerate chemical reactions with slow kinetics or bad mass transport, which is usually hard to improve inside the electrochemical cell. As a consequence of low acid–base crossover, high current efficiency and decoupling the application of acid–base from the electrochemical cell, the acid–base generator for carbon management is compatible with intermittent renewable energy sources. We also suggest that for future works, parameters like current density, energy cost, current efficiency of CO_2 capture, CO_2 source concentration and CO_2 outgas concentration, if not decoupled, should be reported and optimized together in a cycled or steady-state operation. The generator can be further adapted and used for applications like electrochemical mining, metal recycling and battery recycling.

Materials and synthesis

Acetic acid (HOAc), sodium acetate (NaOAc), poly(acrylic acid) sodium salt (NaPAA, average MW 2000), potassium hydrogen phthalate, NaCl, NaBr, NaOH, 5% Ir/C were purchased from VWR International. (((9,10-Dioxo-9,10-dihydroanthracene-2,6-diyl)bis(oxy))bis(propane-3,1-diyl))bis(phosphonic acid) (2,6-DPPEAQ) was purchased directly from TCI Chemicals. Fumasep® E-610K and Nafion® 212 as cation exchange membrane, Selemion® DSV-N anion exchange membrane, and Fumasep® FAB-PK-75 bipolar membrane were purchased and only soaked in 1 M NaCl or KCl solution before usage.

Electrodes for aqueous electrolytes were carbon papers (SGL 39AA) baked at 400 °C overnight. Center chambers for three-chamber cells were made by 3-D printing (see Fig. S24, ESI†). The center chamber thickness is 0.2 or 1.0 mm.

NaOAc electrolyte was prepared by dissolving solid NaOAc in deionized water. NaOH solution for titration was prepared by dissolving 2 g NaOH weighed with an analytical balance in a 500 mL volumetric flask. HCl solution for titration was prepared by diluting 35% concentrated HCl in a 500 mL volumetric flask to obtain approximately 0.1 M HCl. Then we titrated the diluted HCl with standard NaOH solution prepared previously to get the accurate concentration of the diluted HCl. Analytical pure potassium hydrogen phthalate was used as standard to calibrate the concentration of the prepared base.

Pt–Ru catalysts for the hydrogen evolution reaction were prepared by sputtering 30 nm PtRu alloy with an atomic ratio of 7 to 3 on carbon papers using a multi-target magnetron sputtering system. The GDE (0.5 mg cm^{-2} 60% platinum on vulcan – carbon paper electrode) for the hydrogen oxidation reaction was ordered from Fuel Cell Store.

pH measurements were taken using METTLER TOLEDO micro pH probe connected to and recorded by AE channels in a Gamry potentiostat.

Hydroxide crossover flux across the CEM

Hydroxide crossover was evaluated in a standard 5 cm^2 flow cell with two chambers (receiving and source chambers). The receiver chamber contained 40 mL 0.4 M $K_4Fe(CN)_6$, and its initial pH was set at approximately 7. The source chamber contained 200 mL NaOH, and its concentration varied in different groups (0.5 M, 1 M or 2 M). The two chambers were separated by a cation exchange membrane (E610-K). During each crossover test, a pH sensor connected to a Gamry electrochemical work station was placed in the anolyte. The current density varied in different groups (100 mA cm^{-1} or 200 mA cm^{-1}). During the test, the pH change was recorded by the Gamry.

Proton crossover flux across in the three-chamber cell

Proton crossover was evaluated in a standard 5 cm^2 flow cell with three chambers. The cathode chamber contained 40 mL 0.5 M KI_3 – KI or $K_3Fe(CN)_6$ – $K_4Fe(CN)_6$ as non-proton-coupled redox couples, and its initial pH was set at approximately 7. The center chamber contained 200 mL 2 M NaOAc/HOAc. The ratio of NaOAc and HOAc varied in different groups (pure NaOAc or 1 : 1). During each crossover test, H_2 was fed into the anode chamber, through a zero gap GDE with HOR catalysts (Pt) as proton source. A pH sensor connected to a Gamry electrochemical work station was placed in the catholyte. The current density varied in different groups. In the test where protons were generated, the pH change was recorded by the Gamry AE channels connecting to the pH probe. Then the pH change was converted into proton concentration change to quantify the proton crossover.

Acid–base generator driven by HER and HOR

Acid–base generation was constructed based on a 5 cm^2 flow cell. Hydrogen was fed into the anode chamber. The center chamber and anode chamber were separated by one Nafion-212. 20 mL 2 M NaOAc was cycled as center chamber electrolyte, pumped at 53 mL min^{-1}. The center chamber and the cathode chamber were separated by one E-610K. 20 mL 0.1 M NaCl was cycled in the cathode chamber, pumped at 53 mL min^{-1}. For the demonstration of real applications, DI water or 2 M NaOAc was used as electrolytes cycled in the cathode chamber. The initial composition of the catholyte does not significantly influence the current efficiency. The cell was operated with a constant current (20 mA cm^{-2}, 50 mA cm^{-2}, 100 mA cm^{-2}, 200 mA cm^{-2}) by using a Biologic. When applying a constant current of 200 mA cm^{-2}, the pump and current applied were stopped every 10 minutes. Then we used a pipette to take out 1 mL solution from the cathode solution, and used the standard HCl to titrate the solution taken out, thus the volume and concentration of the cathode solution at each time point during the operation to take out solution can be determined. By multiplying the volume and concentration, we can obtain the total amount of NaOH in the cathode solution, and by subtracting the amount of NaOH in the cathode solution at two adjacent time points from each other, we can know the increased molarity of NaOH during these 10 minutes. The theoretical amount of increased molarity NaOH in 10 minutes can be calculated by multiplying the current with time and dividing the result by the faradaic constant. Finally, by dividing the amount of newly generated NaOH measured in the experiment by the theoretical amount, we can get the short time current efficiency in every 10 minutes, denoted as differential current efficiency. For current density of 20 mA cm^{-2}, 50 mA cm^{-2} and 100 mA cm^{-2}, the time period between two operations to take out solution is 100 minutes, 40 minutes and 20 minutes respectively to ensure that the theoretical amount of newly generated NaOH between two time points under different current densities is the same as the 200 mA cm^{-2} experiment, so that the differential current efficiency is comparable. The integrated current efficiency during the whole experiment can be calculated by integrating differential current efficiency of each short time period. The corresponding integrated energy cost of each accumulated concentrations of acid and base was calculated by integrating cell voltage as a function of capacity passing through the circuit, and multiplying by the integrated current efficiency.

Cell polarization

To determine the high-frequency area-specific resistance (ASR), electrochemical impedance spectroscopy (EIS) measurements were conducted with a perturbation of 10 mV and frequencies ranging from 1 to 100 000 Hz. The battery's open-circuit voltage (OCV) was also measured. Additionally, the potential was swept from 0.0 to 2.0 V at a scan rate of 100 mV s^{-1}, and the current response was measured to create polarization curves.

Acid–base generator powered by alternative sources

The cell structure and electrolytes of the acid–base generator were the same as the generator driven by HER and HOR. To realize looped hydrogen in the cell, a gas bag was filled with H_2 in advance. The hydrogen gas bag was connected to the anode chamber. The initial H_2 was pushed through the cell to remove the remaining air in the cell. Then the gas outlet was immediately closed, with only H_2 gas in the cell system to initiate the reaction.

To power the generator by solar cells, the cathode and anode were connected to a solar cell kit (SUNYIMA 10pcs Mini Mono-crystalline Solar Cells Solar System Kit 50 mm × 50 mm/1.96″ × 1.96″ 2V 160MA), where two solar cells were connected in series as a group and three groups were connected in parallel to power the generator. To power the generator by flow batteries, the cathode and anode were connected to a 2.5 cm^2 flow battery using three pieces of baked SGL-39AA as electrodes for each side. 40 mL 1 M pH 4 KBr was used as the posolyte and 40 mL pH 12 0.2 M DPPEAQ supported by 1 M NaCl was used as negolyte (capacity limiting side). The flow battery was first charged under constant current of 40 mA cm^{-2} to 90% state of charge. Then the Br_2 side was connected to the anode of the generator and the DPPEAQ side was connected to the cathode of the generator. Two multimeters were connected to the generator to measure the voltage and current. After the cell operation, we integrated the cell current against time, then divided the result by Faraday's constant to calculate the theoretical molarity of generated NaOH. The accumulated NaOH was titrated, and the current efficiency was calculated accordingly.

BPM based acid–base generator

The experiment was constructed based on a 5 cm^2 flow cell, using four chambers and three membranes. The cathode chamber and anode chamber shared 40 mL 0.5 M $K_4Fe(CN)_6$/$K_3Fe(CN)_6$. 20 mL 2 M NaOAc was cycled in chamber that faces the cation-exchange layer of the bipolar membrane, and this chamber and the cathode chamber were separated by one E-610K CEM. 20 mL 0.1 M NaCl was cycled in chamber that faces the anion-exchange layer of the bipolar membrane, and this chamber and the anode chamber were separated by another E-610K CEM. All solutions were pumped at 53 mL min^{-1}. The cell was operated with a constant current (20 mA cm^{-2}, 50 mA cm^{-2}, 100 mA cm^{-2}, 200 mA cm^{-2}) by using a Biologic. The generated NaOH was collected in the chamber between the anode chamber and the anion-exchange layer of the bipolar membrane, and titrated to calculate integrated current efficiency.

Acid–base generator driven by OER and ORR

The experiment was constructed based on a 5 cm^2 flow cell. It had three chambers and two membranes. Air was pumped into the cathode chamber using GDE coated with ORR catalysts.

20 mL 0.1 M NaCl 20 mL was cycled as center chamber electrolyte, pumped at 53 mL min^{-1}. The center chamber and the cathode chamber were separated by one DSV-N AEM. 20 mL 2 M NaOAc was cycled in the anode chamber, which used a 5 mg cm^{-2} Ir/C coated electrode for OER. The anode solution was pumped at 53 mL min^{-1}. The cell was operated with a constant current (20 mA cm^{-2}, 50 mA cm^{-2}, 100 mA cm^{-2}) by using a Biologic the generated NaOH was collected in the center chamber and titrated to calculate integrated current efficiency.

Acid–base generator driven by water splitting

The experiment was constructed based on a 5 cm^2 flow cell. It had two chambers and one membrane. 20 mL 0.1 M NaCl was cycled as the cathode chamber solution, pumped at 53 mL min^{-1}. The cathode chamber with carbon paper coated by PtRu as HER electrodes and the anode chamber with carbon paper coated by Ir/C as OER electrodes were separated by one E-610K CEM. 20 mL 2 M NaOAc was cycled in the anode chamber. The anode solution was pumped at 53 mL min^{-1}. The cell was operated with a constant current (20 mA cm^{-2}, 50 mA cm^{-2}, 100 mA cm^{-2}, 200 mA cm^{-2}) by using a Biologic. The catholyte was collected and titrated to calculate integrated current efficiency.

Author contributions

D. X. conceived the idea. D. X., and Z. Y. designed and conducted cell tests and electrochemical experiments. Z. Y. and P. Z. tested the crossover of ions. M. S. E. conducted the modelling and simulation. M. J. A. supervised the project. All authors drafted and edited the manuscript.

Data availability

The data supporting this article have been included as part of the ESI.† Codes created for this article can be found on Github: **https://github.com/memanuel/acid-base-gen/tree/main**.

Conflicts of interest

A patent application related to the work has been filed.

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References

1 C. Breyer, M. Fasihi, C. Bajamundi and F. Creutzig, Direct air capture of CO2: a key technology for ambitious climate change mitigation, *Joule*, 2019, **3**(9), 2053–2057.
2 A. Prajapati, R. Sartape, M. T. Galante, J. Xie, S. L. Leung, I. Bessa, M. H. Andrade, R. T. Somich, M. V. Rebouças and G. T. Hutras, Fully-integrated electrochemical system that captures CO_2 from flue gas to produce value-added chemicals at ambient conditions, *Energy Environ. Sci.*, 2022, **15**(12), 5105–5117.
3 K. S. Lackner, A guide to CO2 sequestration, *Science*, 2003, **300**(5626), 1677–1678.
4 S. P. Singh, P. Hao, X. Liu, C. Wei, W. Q. Xu, N. Wei, X. Li, H. Lu and A. Y. Ku, Large-scale affordable CO2 capture is possible by 2030, *Joule*, 2019, **3**(9), 2154–2164.
5 Z. H. Liang, W. Rongwong, H. Liu, K. Fu, H. Gao, F. Cao, R. Zhang, T. Sema, A. Henni, K. Sumon, D. Nath, D. Gelowitz, W. Srisang, C. Saiwan, A. Benamor, M. Al-Marri, H. Shi, T. Supap, C. Chan, Q. Zhou, M. Abu-Zahra, M. Wilson, W. Olson, R. Idem and P. P. Tontiwachwuthikul, Recent progress and new developments in post-combustion carbon-capture technology with amine based solvents, *Int. J. Greenhouse Gas Control*, 2015, **40**, 26–54.
6 W. H. Lee, X. Zhang, S. Banerjee, C. W. Jones, M. J. Realff and R. P. Lively, Sorbent-coated carbon fibers for direct air capture using electrically driven temperature swing adsorption, *Joule*, 2023, **7**(6), 1241–1259.
7 T. Wang, K. S. Lackner and A. Wright, Moisture swing sorbent for carbon dioxide capture from ambient air, *Environ. Sci. Technol.*, 2011, **45**(15), 6670–6675.
8 E. J. Kim, R. L. Siegelman, H. Z. Jiang, A. C. Forse, J.-H. Lee, J. D. Martell, P. J. Milner, J. M. Falkowski, J. B. Neaton, J. A. Reimer, S. C. Weston and J. R. Long, Cooperative carbon capture and steam regeneration with tetraamine-appended metal–organic frameworks, *Science*, 2020, **369**(6502), 392–396.
9 R. L. Siegelman, E. J. Kim and J. R. Long, Porous materials for carbon dioxide separations, *Nat. Mater.*, 2021, **20**(8), 1060–1072.
10 W. Zhang, E. Gao, Y. Li, M. T. Bernards, Y. He and Y. Shi, CO2 capture with polyamine-based protic ionic liquid functionalized mesoporous silica, *J. CO2 Util.*, 2019, **34**, 606–615.
11 X. Shi, H. Xiao, H. Azarabadi, J. Song, X. Wu, X. Chen and K. S. Lackner, Sorbents for the direct capture of CO2 from ambient air, *Angew. Chem., Int. Ed.*, 2020, **59**(18), 6984–7006.
12 D. W. Keith, G. Holmes, D. S. Angelo and K. Heidel, A process for capturing CO2 from the atmosphere, *Joule*, 2018, **2**(8), 1573–1594.
13 N. Mac Dowell, D. M. Reiner and R. S. Haszeldine, Comparing approaches for carbon dioxide removal, *Joule*, 2022, **6**(10), 2233–2239.
14 K. M. Diederichsen, R. Sharifian, J. S. Kang, Y. Liu, S. Kim, B. M. Gallant, D. Vermaas and T. A. Hatton, Electrochemical methods for carbon dioxide separations, *Nat. Rev. Methods Primers*, 2022, **2**(1), 68.

15 M. C. Stern, F. Simeon, H. Herzog and T. A. Hatton, Post-combustion carbon dioxide capture using electrochemically mediated amine regeneration, *Energy Environ. Sci.*, 2013, **6**(8), 2505–2517.
16 M. Wang, H. J. Herzog and T. A. Hatton, CO2 capture using electrochemically mediated amine regeneration, *Ind. Eng. Chem. Res.*, 2020, **59**(15), 7087–7096.
17 G. Lee, Y. C. Li, J.-Y. Kim, T. Peng, D.-H. Nam, A. Sedighian Rasouli, F. Li, M. Luo, A. H. Ip and Y.-C. Joo, Electrochemical upgrade of CO2 from amine capture solution, *Nat. Energy*, 2021, **6**(1), 46–53.
18 H. Seo, M. Rahimi and T. A. Hatton, Electrochemical carbon dioxide capture and release with a redox-active amine, *J. Am. Chem. Soc.*, 2022, **144**(5), 2164–2170.
19 X. Li, X. Zhao, Y. Liu, T. A. Hatton and Y. Liu, Redox-tunable Lewis bases for electrochemical carbon dioxide capture, *Nat. Energy*, 2022, **7**(11), 1065–1075.
20 Y. Liu, H.-Z. Ye, K. M. Diederichsen, T. Van Voorhis and T. A. Hatton, Electrochemically mediated carbon dioxide separation with quinone chemistry in salt-concentrated aqueous media, *Nat. Commun.*, 2020, **11**(1), 2278.
21 K. M. Diederichsen, Y. Liu, N. Ozbek, H. Seo and T. A. Hatton, Toward solvent-free continuous-flow electrochemically mediated carbon capture with high-concentration liquid quinone chemistry, *Joule*, 2022, **6**(1), 221–239.
22 J. M. Barlow, L. E. Clarke, Z. Zhang, D. Bím, K. M. Ripley, A. Zito, F. R. Brushett, A. N. Alexandrova and J. Y. Yang, Molecular design of redox carriers for electrochemical CO2 capture and concentration, *Chem. Soc. Rev.*, 2022, **51**(20), 8415–8433.
23 S. Jin, M. Wu, R. G. Gordon, M. J. Aziz and D. G. Kwabi, pH swing cycle for CO 2 capture electrochemically driven through proton-coupled electron transfer, *Energy Environ. Sci.*, 2020, **13**(10), 3706–3722.
24 S. Pang, S. Jin, F. Yang, M. Alberts, L. Li, D. Xi, R. G. Gordon, P. Wang, M. J. Aziz and Y. Ji, A phenazine-based high-capacity and high-stability electrochemical CO2 capture cell with coupled electricity storage, *Nat. Energy*, 2023, **8**(10), 1126–1136.
25 S. Jin, M. Wu, Y. Jing, R. G. Gordon and M. J. Aziz, Low energy carbon capture *via* electrochemically induced pH swing with electrochemical rebalancing, *Nat. Commun.*, 2022, **13**(1), 2140.
26 M. Hou, L. Chen, Z. Guo, X. Dong, Y. Wang and Y. Xia, A clean and membrane-free chlor-alkali process with decoupled Cl2 and H2/NaOH production, *Nat. Commun.*, 2018, **9**(1), 438.
27 J. Yang, W.-H. Li, H.-T. Tang, Y.-M. Pan, D. Wang and Y. Li, CO2-mediated organocatalytic chlorine evolution under industrial conditions, *Nature*, 2023, **617**(7961), 519–523.
28 S. M. Davis, G. E. Gray and P. A. Kohl, Candidate membranes for the electrochemical salt-splitting of Sodium Sulfate, *J. Appl. Electrochem.*, 2008, **38**, 777–783.
29 G. H. Rau, Electrochemical splitting of calcium carbonate to increase solution alkalinity: Implications for mitigation of carbon dioxide and ocean acidity, *Environ. Sci. Technol.*, 2008, **42**(23), 8935–8940.
30 C. Zhou, J. Ni, H. Chen and X. Guan, Harnessing electrochemical pH gradient for direct air capture with hydrogen and oxygen by-products in a calcium-based loop, *Sustainable Energy Fuels*, 2021, **5**(17), 4355–4367.
31 J. Zhou and X. Guan, Closed-Loop and Precipitation-Free CO2 Capture Process Enabled by Electrochemical pH Gradient, *ChemSusChem*, 2025, e202401533.
32 Y. Xu, S. Liu, J. P. Edwards, Y. C. Xiao, Y. Zhao, R. K. Miao, M. Fan, Y. Chen, J. E. Huang, E. H. Sargent and D. Sinton, Regeneration of direct air CO2 capture liquid via alternating electrocatalysis, *Joule*, 2023, **7**(9), 2107–2117.
33 I. A. Digdaya, I. Sullivan, M. Lin, L. Han, W.-H. Cheng, H. A. Atwater and C. Xiang, A direct coupled electrochemical system for capture and conversion of CO2 from oceanwater, *Nat. Commun.*, 2020, **11**(1), 4412.
34 M. Rahimi, G. Catalini, S. Hariharan, M. Wang, M. Puccini and T. A. Hatton, Carbon Dioxide Capture Using an Electrochemically Driven Proton Concentration Process, *Cell Rep. Phys. Sci.*, 2020, **1**(4), 100033.
35 P. Zhu, Z.-Y. Wu, A. Elgazzar, C. Dong, T.-U. Wi, F.-Y. Chen, Y. Xia, Y. Feng, M. Shakouri, J. Y. Kim, Z. Fang, T. A. Hatton and H. Wang, Continuous carbon capture in an electrochemical solid-electrolyte reactor, *Nature*, 2023, **618**(7967), 959–966.
36 R. J. Gilliam, B. K. Boggs, V. Decker, M. A. Kostowskyj, S. Gorer, T. A. Albrecht, J. D. Way, D. W. Kirk and A. J. Bard, Low voltage electrochemical process for direct carbon dioxide sequestration, *J. Electrochem. Soc.*, 2012, **159**(5), B627.
37 L. Yan, J. Bao, Y. Shao and W. Wang, An electrochemical hydrogen-looping system for low-cost CO2 capture from seawater, *ACS Energy Lett.*, 2022, **7**(6), 1947–1952.
38 X. Zhang, Z. Fang, P. Zhu, Y. Xia and H. Wang, Electrochemical regeneration of high-purity CO2 from (bi) carbonates in a porous solid electrolyte reactor for efficient carbon capture, *Nat. Energy*, 2024, 1–11.
39 Z. J. Schiffer, É. Lucas, N. B. Watkins, S. Ardo, C. Xiang and H. A. Atwater, Acid and base generation *via* an electrochemical hydrogen-looping cell tailored for carbon removal applications, *Device*, 2024, **2**(11), 100506.
40 B. Charnay, Y. Chen, R. Agarwal, J. Misleh, G. Wright, E. Sauve, W. L. Toh, Y. Surendranath and M. Kanan, Membrane-Free Electrochemical Production of Acid and Base Solutions Capable of Processing Ultramafic Rocks, *ChemRxiv*, 2023, preprint, DOI: 10.26434/chemrxiv-2023-5tndz.
41 K. Nagasubramanian, F. Chlanda and K.-J. Liu, Use of bipolar membranes for generation of acid and base—an engineering and economic analysis, *J. Membr. Sci.*, 1977, **2**, 109–124.
42 C. T. Supuran, Carbonic anhydrases-an overview, *Curr. Pharm. Des.*, 2008, **14**(7), 603–614.
43 L. D. Ellis, A. F. Badel, M. L. Chiang, R. J.-Y. Park and Y.-M. Chiang, Toward electrochemical synthesis of cement—An electrolyzer-based process for decarbonating CaCO3 while producing useful gas streams, *Proc. Natl. Acad. Sci. U. S. A.*, 2020, **117**(23), 12584–12591.

44 B. Huskinson, M. P. Marshak, C. Suh, S. Er, M. R. Gerhardt, C. J. Galvin, X. Chen, A. Aspuru-Guzik, R. G. Gordon and M. J. Aziz, A metal-free organic-inorganic aqueous flow battery, *Nature*, 2014, **505**(7482), 195–198.
45 D. Xi, Z. Yang, A. M. Alfaraidi, Y. Jing, R. G. Gordon and M. J. Aziz, Single-membrane pH-decoupling aqueous batteries using proton-coupled electrochemistry for pH recovery, *Energy Adv.*, 2024, **3**(8), 1911–1918.
46 D. Xi, A. M. Alfaraidi, J. Gao, T. Cochard, L. C. I. Faria, Z. Yang, T. Y. George, T. Wang, R. G. Gordon, R. Y. Liu and M. J. Aziz, Mild pH-decoupling aqueous flow battery with practical pH recovery, *Nat. Energy*, 2024, **9**(4), 479–490.
47 H. Q. Dinh, W. L. Toh, A. T. Chu and Y. Surendranath, Neutralization short-circuiting with weak electrolytes erodes the efficiency of bipolar membranes, *ACS Appl. Mater. Interfaces*, 2023, **15**(3), 4001–4010.